

You can't "prove" a specific Indian burial ground from soil alone, but you can absolutely build an app that operationalizes a **forensic** workflow: integrating soil chemistry, geophysics, records, and tribal/archaeological input into probabilistic evidence of sensitive burials at 17642 Beach Blvd / historic Wintersburg lowlands. [keele-repository.worktribe](<https://keele-repository.worktribe.com/output/409990/the-detection-of-forensic-and-archaeological-burials-using-geophysics-and-soil-analysis>)

1. Scientific basis: soil + burials

Key technical pillars you can lean on:

- Cadaver decomposition measurably alters local soil geochemistry (e.g., elevated phosphate, potassium, sulphate, sodium, conductivity, pH shifts) compared with background soils. [keele-repository.worktribe](<https://keele-repository.worktribe.com/output/409990/the-detection-of-forensic-and-archaeological-burials-using-geophysics-and-soil-analysis>)
- Multi-method geophysics (GPR, electrical resistivity, EM induction, magnetic susceptibility) can detect old burials centuries after interment when soil conditions are favorable. [keele-repository.worktribe](<https://keele-repository.worktribe.com/output/409990/the-detection-of-forensic-and-archaeological-burials-using-geophysics-and-soil-analysis>)
- New mobile soil labs (e.g., Subterra Grey) show it is feasible to run in situ spectroscopy and soil compaction tests to flag potential graves and unmarked burials non-invasively. [s4laboratories](<https://www.s4laboratories.com/post/reclaiming-lost-history-with-technology>)

Your app should never claim "legal proof" from soil alone; instead it should score **likelihood** and flag "Areas of Potential Burial Sensitivity (APBS)" for further tribal and archaeological review.

2. Data model for this specific site

Design the data schema around your Wintersburg / Navigation Center fact pattern:

- Parcels and history
 - Parcel table keyed by APN (167-472-09), historic addresses (17631 Cameron Lane, 17642 Beach Blvd), tract (Tract 405, Lot 7), and time-bounded ownership/land use states (agricultural, industrial, shelter, future affordable housing). [latimes](<https://www.latimes.com/socal/daily-pilot/news/story/2020-11-03/huntington-beach-holds-ribbon-cutting-for-new-homeless-navigation-center>)
- Cadastral and infrastructure layer
 - 1947 Survey No. 5403 digitized into GIS: point and polygon features for "Well-pump" (3x3 ft concrete base), "Pump Ho." (15.6 ft dimension), 8x3 ft iron tanks on 3-in concrete piers, offsets from the Cameron-parallel "base" line within bounds (Newman, Cameron, Huntington

Beach Blvd, Vane). [keele-repository.worktribe] (<https://keele-repository.worktribe.com/output/409990/the-detection-of-forensic-and-archaeological-burials-using-geophysics-and-soil-analysis>)

- Store explicit coordinate transforms from handwritten offsets to modern EPSG:4326 / local projected CRS.
 - Geophysics / EM induction layer
 - For each survey (GPR, EM, resistivity, mag): raster or point grids with amplitude/value + confidence + depth window, tagged with acquisition parameters.
 - Soil chemistry / spectroscopy layer
 - Points for each probe/sample: depth profile; major ions (phosphate, potassium, sodium, sulphate), conductivity, pH, organic matter, plus full spectra (VIS-NIR) if collected.
- [s4laboratories] (<https://www.s4laboratories.com/post/reclaiming-lost-history-with-technology>)
- Cultural/archaeological context
 - Polygon for Wintersburg lowlands; references to known Gabrielino-Tongva cultural landscapes and the tribe's broader ancestral footprint in the LA Basin.
- [gabrielinotongva] (<https://gabrielinotongva.org>)
- Linked citations to regional site records, NAHC Sacred Lands File (once obtained), and any Phase I/II cultural resources reports.

3. Core app workflows and UX

Think in terms of 4 main modules; you can ship an MVP with 1-3 and stub 4.

A. Field data capture (mobile/web)

- Map view: basemap + parcel boundary + digitized 1947 survey features overlaid on current Navigation Center footprint.
- [youtube] (https://www.youtube.com/watch?v=IVZa0n5EQ_E)
- Guided sampling grid: user draws AOIs; app generates systematic or stratified grid (e.g., 1-2 m spacing) that avoids known concrete pads/well infrastructure.
 - For each point: form to enter
 - GPS location, depth, instrument used (Subterra-like probe vs auger), soil moisture, land cover.
 - Auto-import readings from compatible probes (conductivity, spectra, pH, insertion pressure).
 - Immediate feedback: simple "traffic light" indicator (green/amber/red) based on deviation from site-specific background chemistry and geophysics.

B. Analytics engine (server or on-device)

Implement a "Burial Likelihood Index (BLI)" per cell:

- Inputs per cell:
 - Soil chemistry deltas from local background (z-scores for phosphate, K, Na, SO₄, conductivity, pH). [keele-repository.worktribe] (<https://keele-repository.worktribe.com/output/409990/the-detection-of-forensic-and-archaeological-burials-using-geophysics-and-soil-analysis>)

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- Geophysics anomalies (GPR hyperbolas, resistivity lows/highs, EM conductivity spikes) with depth consistent with expected historic burials. [keele-repository.worktribe] (<https://keele-repository.worktribe.com/output/409990/the-detection-of-forensic-and-archaeological-burials-using-geophysics-and-soil-analysis>)

- Stratigraphy/compaction anomalies from probe pressure data (looser backfill pockets vs undisturbed matrix).

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- Land-use constraints (distance from historic well pad/pump house/tanks, to avoid misclassifying industrial features).

- Model approach:

- Phase 1: expert-defined rule engine (e.g., high phosphate + high conductivity + GPR anomaly + no metal = "High BLI").

- Phase 2: if you can assemble training data from other validated burial vs control sites, add a supervised ML classifier (e.g., gradient boosting) that outputs probability bands, not binary calls.

Outputs:

- Raster heatmap of BLI over the parcel (0-1 or 0-100 scale).

- For each AOI, printable technical memo summarizing methods, inputs, and BLI distribution for submission to tribes, city, and agencies.

C. Provenance and legal defensibility

Because you're using this in a contested, high-stakes context:

- Full audit log: keep hash-chained records of each reading, its raw value, device ID, operator, timestamp, and any transformations applied.

- Versioned models: every scoring run is tied to a particular ruleset/model version; report includes that version id and parameter set.

- Evidence bundle export:

- GIS (GeoPackage/GeoJSON) containing BLI raster, sampling points, and overlays.

- PDF narrative that explains, in plain terms, that BLI is a **screening** tool and that final determinations rest with qualified archaeologists and tribal monitors.

D. Archaeological & tribal review portal (later phase)

- Secure user roles for tribal governments, archaeologists, city staff, and possibly community advocates.

- Ability to annotate hotspots, request additional sampling, or flag culturally sensitive zones that should never be ground-truthed invasively.

- Integration hook for NAHC and SHPO case numbers so your evidence aligns with regulatory processes (AB 52, CEQA cultural resources).

4. Tech stack and architecture

A concrete, testable stack you could implement:

- Frontend
 - Web: React + Mapbox GL or Leaflet for mapping; offline-capable PWA if you want field usability.
 - Mobile: React Native or Flutter wrapper sharing logic with the web client.
- Backend
 - API: FastAPI (Python) or Node/Express for REST or GraphQL.
 - Data and spatial: Postgres + PostGIS for parcels, survey geometry, sampling points, rasters.
 - Analytics: Python stack (NumPy, SciPy, scikit-learn) to compute BLI rasters; store outputs back into PostGIS.
- Integration
 - Ingest from soil probes/mobile labs via BLE/serial → small gateway service that normalizes device data schema (start with manual CSV upload, then automate).
 - [s4laboratories] (<https://www.s4laboratories.com/post/reclaiming-lost-history-with-technology>)
 - Use GDAL/OGR for importing 1947 survey scans that have been georeferenced or for warping historic air photos into your base map.

Security/ethics:

- Implement strong ACLs and encryption at rest; location of likely burial grounds is extremely sensitive, especially for Gabrielino-Tongva and other tribal communities in the LA Basin.
- [nahc.ca] (<https://nahc.ca.gov/cp/tribal-atlas-pages/gabrielino-tongva-nation/>)
- Bake in governance: default that tribes control visibility of high-likelihood areas outside required regulatory audiences.

5. How to move from concept to proof-of-concept

For a POC tied specifically to 17642 Beach / Wintersburg:

1. Digitize Survey No. 5403 and georeference it into a GIS so you can see the historic well, pump house, and tanks relative to the modern Navigation Center lot lines and buildings.
- [latimes] (<https://www.latimes.com/socal/daily-pilot/news/story/2020-11-03/huntington-beach-holds-ribbon-cutting-for-new-homeless-navigation-center>)
2. Implement simple web map viewer showing: parcel, 1947 features, current shelter layout (from city/press imagery) as the "context" layer.
- [youtube] (https://www.youtube.com/watch?v=IVZa0n5EQ_E)
3. Design a sampling plan that explicitly avoids known concrete/metal infrastructure but covers low-lying areas that, historically, could have been culturally used by Tongva people.
4. Hard-code a simple rule-based BLI: e.g., "if phosphate and conductivity above threshold AND no metal anomaly in EM AND GPR suggests discrete anomaly at 0.8-1.5 m depth, classify as High." [keele-repository.worktribe] (<https://keele-repository.worktribe>)

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5. Run with a ****simulated**** dataset first; once real field partners and tribal consent are in place, swap in live data.

If you'd like, I can next sketch actual Postgres/PostGIS table definitions and a rough JSON API spec for the BLI computation endpoint so you or a contractor can start building against it.